

PORTERVILLE, CA, USA

WMO: 723895

Lat: 36.029N

Lon: 119.063W

Elev: 442

StdP: 14.46

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23149

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	30.2	33.1	20.8	15.9	40.8	24.8	19.2	43.3	15.3	52.3	12.3	52.3	3.1	110	0.428

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.4	101.7	70.8	99.5	69.5	97.4	68.4	73.3	96.9	71.5	94.7	70.1	93.4	9.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.2	91.4	86.4	62.9	87.3	85.1	61.0	81.6	83.3	37.0	96.6	35.4	95.1	34.2	92.3	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.9	11.5	10.4	DB	27.4	106.4	2.6	2.1	25.6	107.9	24.1	109.1	22.7	110.3	20.8	111.8
			WB	26.1	75.2	2.1	2.3	24.6	76.9	23.4	78.3	22.3	79.6	20.8	81.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	47.8	50.9	55.6	60.3	67.7	75.4	80.9	79.3	74.6	64.5	54.0	47.3	(6)
	DBStd	13.13	4.97	4.91	5.72	6.87	6.91	6.54	4.80	4.53	5.87	6.09	5.63	5.16	(7)
	HDD50	309	104	43	15	4	0	0	0	0	0	0	25	118	(8)
	HDD65	2423	535	395	296	174	49	5	0	0	4	83	332	550	(9)
	CDD50	5148	35	68	188	313	550	763	958	908	739	449	144	33	(10)
	CDD65	1786	0	0	4	33	134	318	493	443	293	67	1	0	(11)
	CDH74	25484	0	6	140	671	2120	4701	7035	5997	3758	1003	53	0	(12)
	CDH80	14120	0	0	23	227	981	2680	4286	3558	2027	334	4	0	(13)
(14) Wind	WSAvg	5.0	3.8	4.3	4.8	5.5	6.0	6.2	6.0	5.7	5.2	4.6	3.9	3.8	(14)
(15) Precipitation	PrecAvg	10.7	1.9	1.9	1.8	1.1	0.4	0.1	0.0	0.0	0.2	0.6	1.5	1.5	(15)
	PrecMax	19.4	7.9	6.5	5.3	4.8	2.9	0.5	0.2	0.3	1.5	2.6	3.5	5.6	(16)
	PrecMin	5.2	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	3.6	1.7	1.7	1.3	1.1	0.6	0.1	0.0	0.0	0.4	0.7	1.0	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.7	73.5	82.4	90.8	99.2	104.3	105.8	103.8	100.2	92.8	79.8	68.2	(19)
		MCWB	56.0	58.6	62.2	64.0	67.8	71.9	73.0	72.1	69.7	66.0	60.1	55.4	(20)
	2%	DB	63.7	69.6	78.7	86.1	93.4	100.2	101.9	100.1	97.4	88.0	74.7	63.8	(21)
		MCWB	53.9	57.2	60.7	62.4	65.3	68.8	71.3	69.9	67.9	64.0	58.4	53.3	(22)
	5%	DB	60.9	64.3	73.3	81.8	90.2	97.4	99.6	98.7	94.6	83.5	70.5	60.9	(23)
		MCWB	53.0	54.8	58.4	60.8	63.9	67.5	70.4	69.2	67.3	62.0	56.7	52.1	(24)
	10%	DB	57.1	61.5	70.1	77.1	85.5	93.3	97.3	95.6	91.1	80.6	65.9	57.0	(25)
		MCWB	51.2	53.1	56.8	59.1	62.8	66.2	69.1	68.0	65.9	60.9	54.8	50.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.2	60.8	64.7	67.8	70.1	74.7	75.8	74.3	73.4	68.4	62.5	58.9	(27)
		MCDB	64.0	68.6	77.8	84.5	94.6	99.3	99.5	99.3	95.9	87.4	75.3	64.6	(28)
	2%	WB	56.0	58.7	62.1	64.0	67.3	71.3	73.6	72.1	70.2	65.8	60.2	55.9	(29)
		MCDB	61.1	66.5	75.7	82.3	91.0	95.9	97.3	95.7	92.1	83.2	71.4	60.6	(30)
	5%	WB	54.0	56.4	59.7	61.8	65.2	69.2	72.0	70.6	68.4	63.8	58.2	53.6	(31)
		MCDB	59.0	63.4	71.8	79.3	87.0	93.9	95.3	94.0	91.3	80.0	68.0	58.5	(32)
	10%	WB	52.0	54.1	57.4	59.8	63.3	67.5	70.4	69.2	66.9	61.7	55.9	51.4	(33)
		MCDB	56.7	60.6	67.9	75.5	83.7	91.3	93.5	92.6	89.6	78.2	64.6	56.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	17.7	20.4	24.9	27.5	29.4	32.1	31.4	31.7	30.8	27.4	22.6	18.7	(35)
		MCDBR	23.3	25.6	31.1	33.9	34.4	34.8	32.2	32.9	33.0	32.5	29.4	23.7	(36)
	5% WB	MCWBR	15.6	16.3	16.7	15.7	14.0	13.9	12.7	13.1	13.8	15.0	16.9	15.3	(37)
		MCWBR	19.5	23.1	28.9	31.4	32.8	33.1	30.7	31.2	31.3	28.3	25.6	18.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.352	0.364	0.365	0.396	0.404	0.388	0.422	0.397	0.388	0.391	0.359	0.354	(40)
		taud	2.344	2.321	2.335	2.256	2.262	2.319	2.257	2.301	2.329	2.272	2.336	2.333	(41)
	Ebn at Noon		264	275	286	281	279	283	272	278	275	261	258	253	(42)
		Edh at Noon	30	35	37	43	43	41	43	40	37	36	30	29	(43)
(44) All-Sky Solar Radiation	RadAvg		827	1146	1612	2040	2391	2679	2534	2358	1980	1449	1011	758	(44)
	RadStd		93	125	125	128	135	102	105	74	55	80	51	86	(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (25 neighbors)	+0.92	N/A	-2.03	+0.83	N/A	N/A	N/A	-221	+311	+122	(47)

Nomenclature: See separate page